

PIONIX

phyVERSO

Hardware Revision 1

Technical Specification and Manual

The **phyVERSO-EVCS** is a fully-featured smart controller for EV charging stations in both AC and DC applications manufactured by PHYTEC. On the software side, BaseCamp controls the board's functionality and allows for a great integration experience in your own industry solutions.

BaseCamp is a stable and production-ready wrapper around the open-source operating system **EVerest**.



Process for support and access of further documentation

After having received your board, you will find a postcard with a QR code to a pipedrive-hosted webform. Register your product there and you will receive an email with further information from Pionix. This email will also include the username and password required to access the documentation of Pionix BaseCamp.

Tech Specs:

- Charging interface:
 - Two charging ports for AC (Mode 2 and 3) or DC (Mode 4)
 - Mixed AC/DC configuration possible
 - Compatible with Type 1 and 2 AC / DC CCS connectors and cable assemblies
 - AC up to 43kW, DC up to 1000V/500A (limited by external circuitry only)
 - Solid State Relay (AC) / NMOS switches (DC) to drive contactor for each port
 - Connector for contactor feedback (1x AC / 1x DC)
 - Basic IEC 61851-1 signaling
 - HLC according to DIN SPEC 70121 (DC), ISO 15118-2 (AC/DC), ISO 15118-20 (AC/DC/BPT, available Q4/23)
 - Bidirectional charging support
 - RDC-MD connector (DC ground faults) for each port
 - Connector lock motor driver for each port
 - Backup power supply connector option to enable basic vehicle charging communication and connector-lock unlocking during power loss
 - Several ADC channels for temperature measurement or other use cases (up to 10)
 - Load balancing between the two ports
- Communication interfaces:
 - 2x Gigabit Ethernet
 - Wireless LAN / BLE
 - LTE module
 - 2x PLC GreenPHY ISO 15118-3
 - 2x RS485 / ModBus (1x isolated)
 - 2x CAN FD
 - USB 2.0, SPI, I2C
- Authentication / payment:
 - OCPP 1.6J for backend connection
 - Extensions for OCPP 1.6: security extensions, PnC, Smart Charging
 - OCPP 2.0.1 (available Q4/2023)
 - Plug and Charge support (AC/DC)
 - Autocharge (AC/DC)
 - EIM / RFID reader externally connected
 - Support for German calibration law metering (Eichrecht)

- HMI:
 - Two LVDS display connectors
 - 2x USB 2.0 interface for touch at USB HUB version
- Main Controller:
 - Texas Instruments AM62x based SoM
 - Up to 4 cores, 1 GB-4 GB RAM, 8-32 GB FLASH
 - Supported Yocto BSP with EVerest layer and limited BaseCamp image
- Fan Controller:
 - 4x independent four-wire PWM controlled 12 V fans with tachometer input
 - 1x four-wire PWM controlled 12 V fan without tachometer

The **phyVERSO-EVCS** supports up to two charging connectors and scales from domestic use cases / semi-public scenarios to fast charging parks on the highways. It is mainly targeted for European and North American markets but can be customized for Asian markets.

Disclaimer: *The descriptions and specifications of this product are preliminary and subject to change without further notice. Final product specifications may differ. As PIONIX cannot control or anticipate the conditions under which this product may be used, each user should review the information in the specific context of the planned use. To the maximum extent permitted by law, PIONIX will not be responsible for damages of any nature resulting from the use or reliance upon the information contained in this document.*

Table of Content:

Introduction	5
Disclaimer	6
Safety	7
Technical Specifications	9
Electrical Block Diagram	11
Basic Charging Subsystem Interfaces and Connectors	12
Basic charging communication and motor lock connectors	12
AC connection for PLC and AC contactor / feedback	13
Control and mirror contacts for AC contactor	13
Control and mirror contacts for DC contactor	13
Residual Direct Current Monitoring Device	15
ADC inputs	15
Fan connection	16
Linux System Interfaces and Connectors	18
Wireless LAN/Bluetooth	18
Ethernet	19
RS485	19
Display	19
CAN	20
USB	21
Mini-PCle/LTE	21
Digital input and output	22
Power Line Communication	23
Extension connectors	23
Debugging	24
SOM Fan	25
External power meters	25
RFID reader support	25
Security	25
Certifications	26
Power Supply	26
Input supply	26
Power consumption	27
Backup supply and RTC	27
Temperature Grade	27
Integrating the phyVERSO-EVCS	29
Electrical installation	29
AC Charger Configuration	34

DC Charger Configuration	35
Flashing/Updating the Linux image	35
Flashing via RAUC	35
Flashing via u-boot	36
Accessing the phyVERSO-EVCS via SSH	41
Updating the microcontroller firmware	42
BaseCamp Configuration	43
Appendix	44
List of connectors	44
Revision history	46

Introduction

This manual describes the technical specifications and shows how to integrate the phyVERSO into your own environment.

Disclaimer

This document is drafted for information purposes only and does not constitute a binding offer or contract with PIONIX. PIONIX has compiled this document to the best of its knowledge. No express or implied warranty is given for the completeness, accuracy, reliability, or fitness for the specific purpose of its content and the products and services presented therein. Specifications and performance data contain average values within existing specification tolerances and are subject to change without prior notice. PIONIX explicitly rejects any liability for any direct or indirect damage, in the broadest sense, arising from or related to the use or interpretation of this document.

No part of this document may be modified, reproduced, processed, or distributed in any form or by any means, without the prior written permission of PIONIX.

PIONIX GmbH
Ewald-Renz-Straße 1a
76669 Bad Schönborn
Germany

Safety

WARNING

Not following the installation and user instructions given in this manual will result in the risk of electric shock, which will cause severe injury or death.

Read this manual before working with the phyVERSO board.

Working on electrical installations without using personal protective equipment will result in the risk of injury.

Use personal protective equipment such as eye protection, cut-resistant gloves, and non-slip safety shoes to prevent personal injuries.

WARNING

Not taking precautions against ESD (Electrostatic discharge) can damage electronic components..

Take the necessary precautions against ESD before touching electronic components. Ask your electrician which local requirements apply.

WARNING

Exposure of electronic components to water will result in the risk of electric shock, which will cause severe injury or death.

Cybersecurity note:

This product is designed to be connected to and to communicate information and data via a network interface. It is the owner's sole responsibility to provide and continuously ensure a secure connection between the product and owner's network or any other network (as the case may be).

The owner shall establish and maintain any appropriate measures (such as, but not limited to the installation of firewalls, application of authentication measures, encryption of data, installation of anti-virus programs, etc) to protect the product, the network, its system and the interface against any kind of security breaches,

unauthorized access, interference, intrusion, leakage and/or theft of data or information.

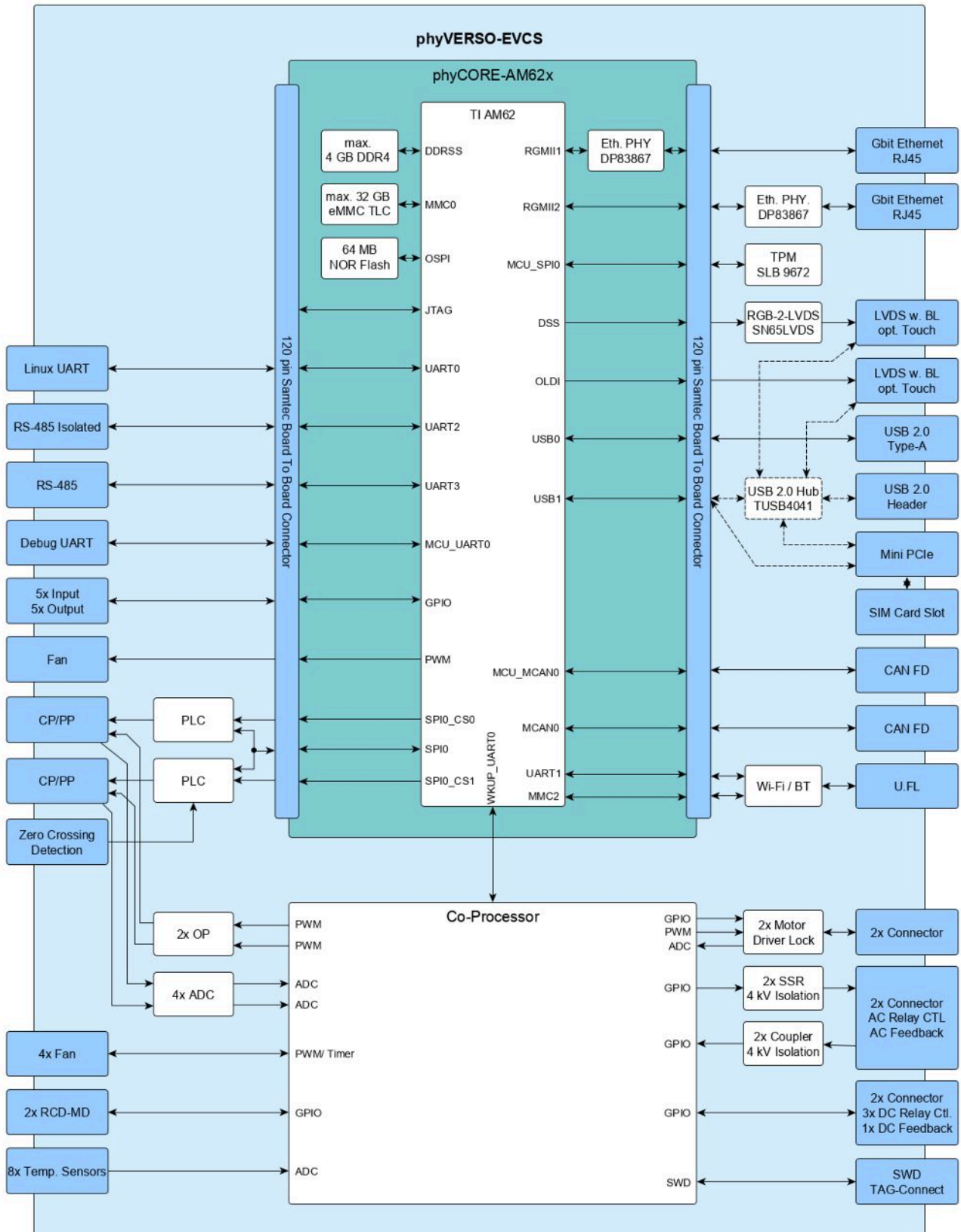
PIONIX is not liable for damages and/or losses related to such security breaches, any unauthorized access, interference, intrusion, leakage and/or theft of data or information.

Technical Specifications

Feature	Description
Electrical properties	
Input supply voltage	12V DC $\pm 5\%$ external supply with double or reinforced isolation with at least 4 kV
Power consumption	Depends on the configuration of the phyVERSO-EVCS and the applied use case. The SOMs power consumption is up to 5 W.
Power cable	WAGO MCS MINI 3.5 mm 3-pol
Maximum charge rate	AC up to 43 kW, DC up to 1000 V/500 A (limited by external circuitry only)
Charging mode	AC (Mode 2 and 3), DC (Mode 4) - mixed AC/DC configuration possible
Charging cable	Depends on charger configuration - Type 1 and 2 AC / DC CCS connectors and cable assemblies
Interface for external AC circuitry	2x SSR with reinforced isolation to drive external contactor with 230 V/55 mA, providing a 5 kV reinforced isolation 2x 3.3 V digital input for relay feedback / mirror contact with reinforced isolation for neutral reference
Interface for external DC circuitry	2x3 NMOS to drive contactor with 12/24 V DC coil 2x 3.3 V digital input for relay feedback / mirror contact
Control Pilot interface	2x ± 12 V PWM signaling according to IEC 61851-1

Connector lock	2x 12 V motor lock driver with a peak current of 1.5 A and ADC feedback for connector lock state
Ground leakage detection	Depends on external RDC module, separate connections for each charging port
Operating temperature range	-40 °C to +70 °C
Connectivity and user interface	
Authorization device	EIM / RFID reader externally connected
Communication standards	2x Gigabit Ethernet Wireless LAN / BLE LTE module 2x CAN FD 2x RS485 / ModBus (1x isolated) USB 2.0 SPI I2C
Powerline Modem	2x PLC GreenPHY ISO 15118-3
Display	2x LVDS display connectors 2x USB 2.0 interface for touch at USB HUB version
Physical information	
Dimensions	120 mm x 210 mm

Electrical Block Diagram



Basic Charging Subsystem Interfaces and Connectors

The basic charging signaling using the CP/PP pins is implemented on a separate microcontroller subsystem. The microcontroller firmware takes care of the electrical safety. Safe switching off is still guaranteed in the event the main Linux system becomes inoperable.

The co-processor is connected via UART to the Linux domain, which allows control from EVerest.

Basic charging communication and motor lock connectors

There are two connectors available for basic charging communication via CP/PP signaling and motor lock feedback and control, one for each charging port. The following table shows the pinout for the basic charging communication connectors (X12/X22):

Pin	Function
1	Motor Driver: OUT1
2	Motor Driver: OUT2
3	Connector Lock: Feedback
4	PE* / "Protective Earth"
5	CP / "Control Pilot"
6	PP / "Proximity Pilot" / "Plug Present"

The Connector Lock Motors are driven by an H-bridge configuration driver so you can also swap pin1/pin2 connections on the motor side for reversing the motor direction in hardware. The Connector Lock feedback is connected to the co-processors ADC with a resistive divider consisting of a 10 k Ω series and 3.3 k Ω resistor to GND.

*Note that only one PE connection should be used, mostly at X1. The PE on X12/X22 should be left open then.

AC connection for PLC and AC contactor / feedback

There is one AC connector available (X9). This is necessary for the zero-crossing detection for the Power Line Communication and to use the AC contactor control and feedback.

The following table shows the pinout for the AC connector (X9):

Pin	Function
1	AC Phase
2	AC Neutral

Control and mirror contacts for AC contactor

There are two connectors available for AC contactor control and feedback, one for each charging port. The following table shows the pinout for the AC contactor control and feedback connectors (X17/X27):

Pin	Function
1	AC contactor feedback (Neutral)
2	AC contactor control (Phase)

Pin 1 of the connector will be measured in reference to the AC phase. Pin 2 will be connected to the AC phase by a solid state relay when activated. The SSR and the optocoupler both provide a reinforced isolation. The AC connector has been described in the last chapter.

Control and mirror contacts for DC contactor

There are two connectors available for DC contactors control and feedback, one for each charging port. The following table shows the pinout for the DC contactor connectors (X15/X25):

Pin	Function
1	12 V / 24 V Input
2	GND
3	Low-Side-Switch Open Drain Output 1
4	Low-Side-Switch Open Drain Output 2
5	Low-Side-Switch Open Drain Output 3
6	GND
7	Contactor mirror contact pin 1 10 k Ω pull-up to 3.3
8	Contactor mirror contact pin 2 feedback input pin to co-processor

Pins 3/4/5 are connected to the drains of low side switching NFETs. Connect your 12 V / 24 V DC coil supply to pin 1 to use the onboard flyback diodes or make sure to connect external protection circuitry against voltage spikes on pins 3/4/5 by the inductive loads of your DC coil contactors.



Caution

Please make sure to connect pin 1 to your DC coil 12 V / 24 V supply to use the integrated flyback diodes or provide external protection circuitry (e.g. flyback diode) against voltage spikes on pins 3/4/5 by inductive loads of the DC coils of your contactors.

There is no isolation provided on these connectors. It is expected that the DC contactors and mirror contacts have reinforced isolation to the power path.

Residual Direct Current Monitoring Device

There are two connectors available for Residual Direct Current Monitoring Devices (RDC-MD), one for each charging port. The following table shows the pinout for the RDC-MD connectors (X14/X24):

Pin	Function
1	RDC-MD: GND
2	RDC-MD: 12 V
3	RDC-MD: Test (12 V)
4	RDC-MD: Fault
5	NTC: ADC (5.1 k Ω Pullup to 3.3 V)
6	NTC: GND

As you can see the RCD-MD connectors also offer NTC/PTC connections to the co-processor ADC with 5.1 k Ω pull-up resistors to 3.3 V.

ADC inputs

The phyVERSO-EVCS offers 10 co-processor ADC inputs.

There are two connectors available for each charging port, offering two temperature sensors per connector. The following table shows the pinout for the temperature sensor connectors (X19/X29):

Pin	Function
1	NTC/PTC-1: ADC/10 k Ω Pullup to 3.3 V
2	NTC/PTC-2: ADC/10 k Ω Pullup to 3.3 V
3	GND

The NTC/PTC connections to the co-processor ADC are equipped with 10 k Ω pull-up resistors to 3.3 V.

There are two connectors available for 5 V parking sensors, one for each charging port. The following table shows the pinout for the parking sensor connectors (X18/X28):

Pin	Function
1	5 V
2	ADC Input with 10 k Ω : 10 k Ω voltage divider to GND
3	GND

Additionally, there are ADC connections for one NTC/PTC connection on each RDC-MD and SPI extension connector ([Extension connectors](#)) as well as Motor Lock feedback and CP/PP connections to the ADC that have been described in the previous respective sections.

Fan connection

There are four connectors available for four-wire PWM controlled 12 V fans. The following table shows the pinout for co-processor controlled fan connectors (X4-X7):

Pin	Function
1	GND
2	12 V
3	Tacho/Sense
4	PWM/Control

All of the fan connectors are fused with a 2.5 A fuse.

As an option, the fan supply can be provided separately at the fan supply connector (X8). The pinout of this connector is shown in the following table:

Pin	Function
1	Fan supply
2	GND

In the case of using an external fan supply you have to change some resistor jumpers. Please contact PHYTEC for additional information on this topic.

Linux System Interfaces and Connectors

As the central control unit for high level charging communication and graphics a phyCORE-AM62x is used. The general features and options are shown in the following table:

phyCORE-AM62x	
Processor	TI AM6232, TI AM6252
Architecture	Dual Core Coretex-A53 / Arm Cortex-M4F up to 400 MHz
Frequency	Up to 1.4 GHz
Graphics	3D GPU - OpenGL 3.x/2.0/1.1, Vulkan 1.2
Memory	
RAM	1 GB default / 4 GB max DDR4
eMMC	8 GB default / 128 GB max
EEPROM	4 KB
SD/MMC Interfaces	3
Display	OLDI/LVDS (4 lanes 2x) and Parallel Display (24 bpp)

In this hardware revision, the phyCORE-AM62x will be connected over two Samtec ASP-205225-01 connectors. As previously mentioned, the SOM is supported by an external co-processor, which is used for the low level charging communication and other safety functionalities.

Wireless LAN/Bluetooth

The onboard 2.4 GHz Wi-Fi 6 and Bluetooth Low Energy Combo wireless module enables the use of Wi-Fi 6 IEEE 802.11 a/b/g/n/ax and BLE 5.2. based on the CC33xx chip from Texas Instruments. A U.FL connector is placed at the combo module for an external antenna connection.

Ethernet

The hardware comes with two ESD protected Gigabit Ethernet ports, accessible over RJ45 connectors (X30/X31). These connectors include a green and a yellow LED each for link state and traffic indication. Both Ethernet Physical Layer Transceivers are connected over RGMII. One is placed on the SOM and one is placed on the baseboard.

For the hardware option with only one Ethernet interface, only the PHY at the SOM is mounted (X30).

RS485

The hardware provides up to two half duplex RS485 interfaces. One non-isolated transceiver (X35) with up to 1 Mbps and one transceiver with basic isolation (X34) and up to 12 Mbps are available. Both interfaces have a 120 Ohm termination and ESD protection.

The following table shows the pinout of both RS485 connectors:

Pin	Function
1	A
2	B
3	GND

In case of the isolated RS485, the ground should be connected to the ground of the communication partner. In case of the non isolated RS485 the ground should only be connected to the communication partner's ground, if that is an isolated ground. This will avoid a ground loop.

Display

The hardware offers two LVDS connectors with 5 differential pairs (4x data and 1x clock) each. One is a native AM62x OLDI (Open LVDS Display Interface) (X60). The other interface comes from a Parallel-to-LVDS converter, which is using the 24 bit RGB-interface of the AM62x as input (X61).

The following table shows the pinout of the connectors X60/X61:

Pin	Function
1, 2, 21, 22	Not connected
3, 7	3.3 V
4, 8, 17, 18, 24, 26, 28, 30	GND
5	LVDS CLK-
6	LVDS CLK+
9	LVDS L0-
10	LVDS L0+
11	LVDS L1-
12	LVDS L1+
13	LVDS L2-
14	LVDS L2+
15	LVDS L3-
16	LVDS L3+
19	Backlight - Enable
20	Backlight - PWM
23, 25, 27, 29	12 V

CAN

The hardware is equipped with up to two CAN-FD transceivers with a maximum data rate of 8 Mbps (X32/X33). Those are connected to the FLEXCAN interface of the AM62x. Both interfaces come with a 120 Ohm termination and ESD protection.

The following table shows the pinout of the connectors X32/X33:

Pin	Function
-----	----------

1	CAN_HIGH
2	CAN_LOW
3	GND

The ground should only be connected to the communication partner's ground, if that is an isolated ground. This will avoid a ground loop.

USB

The AM62x offers two USB 2.0 interfaces. One USB 2.0 interface is available for generic extensions on a Type-A connector. The second interface is routed to an USB 2.0 hub, which has one downstream port connected to the mini-PCle connector (X39), two ports to the display section - where it can be used to connect for example USB-touch-controller (X47/X48) - and one to an expansion connector (X49).

The following table shows the pinout of the connectors X47/X48/X49:

Pin	Function
1	5V
2	DM
3	DP
4	GND

If you have a phyVERSO-EVCS version without USB hub, the second USB 2.0 interface is directly connected to the mini-PCle connector described in the respective section.

Mini-PCle/LTE

The phyVERSO-EVCS has a mini-PCle Connector (X39), that can be used to mount an LTE module. The mini-PCle connector is connected to the AM62x via USB 2.0.

PCle signaling is not available. A SIM socket is provided on the baseboard (X38) connected to the mini-PCle connector. An antenna can be connected to the U.FL connector mounted on the mini-PCle-LTE add-in-card.

Digital input and output

The phyVERSO-EVCS has five 5 V outputs (X44) and five 3.3 V inputs (X43) connected to the AM62x.

The following table shows the pinout of the connector X44:

Pin	Function
1	5 V
2	OUT_5
3, 4	GND
5	OUT_1
6	OUT_4
7	OUT_2
8	OUT_3

The following table shows the pinout of the connector X43:

Pin	Function
1	5 V
2	IN_5
3, 4	GND
5	IN_1
6	IN_4
7	IN_2
8	IN_3

The inputs are 5V-tolerant.

Power Line Communication

The phyVERSO-EVCS features two separate PLC (HomePlug GreenPHY) transceivers, one for each charging port according to ISO 15118-3.

The communication is implemented with two Lumissil CG5317, which are connected via SPI to the Linux system. Both transceivers use the same SPI interface with an individual chip-select-signal each.

Extension connectors

The phyVERSO-EVCS has several extension connectors available.

One SPI interface connector (X10/X20) per charging port. Both connectors are connected to the same interface but have individual Chip Selects. The following table shows the pinout of the connectors X10/X20:

Pin	Function
1	3.3 V
2	CLK
3	nCS
4	MOSI
5	MISO
6	GND
7	NTC: ADC (5.1 k Ω Pullup to 3.3 V)
8	NTC: GND

As you can see the SPI extension connectors also offer NTC/PTC connections to the co-processor ADC with 5.1 k Ω pull-up resistors to 3.3 V.

One I2C interface connector (X11/X21) per charging port. Both connectors are connected to the same interface.. The following table shows the pinout of the connectors X11/X21:

Pin	Function
1	12 V
2	nADD_PWM
3	GND
4	NC
5	3.3 V
6	I2C_SCL
7	I2C_SDA
8	nADD_LS

Debugging

For general communication, a linux console over UART (UART0) can be used (X59). This console can be disabled in software. The pinout for the serial connector X59 is shown in the following table:

Pin	Function
1, 2, 3, 7, 8	not connected
4	UART0_TXD
5	3.3 V output (supplied by phyVERSO), do not connect 3.3 V externally here
6	UART0_RXD
9, 10	GND

Additionally, the UART for the internal M4 core is routed to a connector (X58). This internal M4 Core is currently not used.

SOM Fan

A 12 V fan can be connected to and controlled by the Linux system on the connector X3. The pinout is shown in the following table:

Pin	Function
1	GND
2	12 V
3	not connected
4	PWM/Control

External power meters

External power meters are supported on the RS485/ModBus. BaseCamp supports most available AC/DC power meters. New register definitions can be added easily to the GenericPowerMeter module.

For AC Eichrecht: Bauer BSM power meter drivers are included in open source EVerest. Others can be added easily.

More / new drivers are also available through commercial support from Pionix.

RFID reader support

UART/RS485/SPI/CAN can be used to connect RFID readers.

EVerest supports the widely used PN532 RFID reader IC in the open source edition.

More drivers can be added easily or through commercial support from Pionix.

Security

A Trusted Platform Module (TPM2.0) is connected to the Linux system over SPI, and the phyVERSO-EVCS supports Secure Boot for Linux.

EVERest supports the security white papers for OCPP 1.6 and supports secure connections on OCPP 2.0.1 for securely connecting to the central management system in the cloud. TLS connection to the car via ISO 15118 is supported as well to allow for secure authentication via Plug and Charge from the car to the cloud backend.

For secure storage of private keys for the TLS connections of ISO 15118 and OCPP with TPM and secure updates, contact Pionix (this is not part of the open source edition of EVERest).

A smart card socket for secure key storage is available as a hardware option.

Certifications

The Wi-Fi/BLE module has the following certifications: FCC, IC, CE-RED, Bluetooth SIG.

The phyVERSO-EVCS will be generally tested according to EN IEC 61851-21-2. Since the standard applies to complete charging stations, not all tests of the standard are done for the phyVERSO-EVCS. For example an extra Surge Protection Device is expected, reducing the surge voltage level to 1.5 kV at the AC Connections of the phyVERSO-EVCS.

Power Supply

Input supply

The input voltage is specified in a range of 12 V DC $\pm 5\%$. An external power supply should provide double or reinforced insulation with at least 4 kV.

The following table shows the pinout of the power connector X1:

Pin	Function
1	12V
2	GND
3	PE

Onboard regulators provide 5 V and 3.3 V. For vehicle communication, ± 12 V will be generated from the 3.3 V supply by a boost converter to meet the IEC 61851-1 voltage level requirements.

Power consumption

The power consumption depends on the hardware and software configuration of the phyVERSO-EVCS and the applied use case. The SOMs power consumption is up to 5 W.

Backup supply and RTC

The phyVERSO-EVCS has a 0.47 F supercapacitor mounted which is charged up to 3.3 V when the main power supply is active. The supercapacitor is used to supply an onboard Real-Time-Clock (RTC), which is placed at the SOM.

The RTC has a low power consumption of typically 40 nA at 3 V and keeps the time down to 1.1 V.

For RTC-Backup-Time Calculation, the leakage current of the supercapacitor should be taken into account. It is significantly higher than the RTC current and strongly temperature dependent.

A second onboard backup-supply for the 12 V rail is also available. If the main supply is deactivated, this supply will be used to supply the co-processor, the basic vehicle communication and the motor drivers for the connector lock.

Temperature Grade

The phyVERSO-EVCS is specified for a temperature range of -40 to +70 degree celsius. For the upper temperature limit, the self-heating of the components should be taken into account.

All components are able to work at up to 85 degrees celsius. For most of them, an ambient and PCB maximum temperature of 70 degrees celsius offers enough room for power dissipation.

An exception is the AM62x, which is the most critical part. The AM62x junction temperature range is -40 to +105 degrees celsius, where the clock rate is reduced above 95 degrees celsius and a shutdown is initiated above 100 degrees celsius.

The cooling requirement depends on the load situation and has to be evaluated by the customer.

Integrating the phyVERSO-EVCS

⚠ Caution: Qualified personnel only!

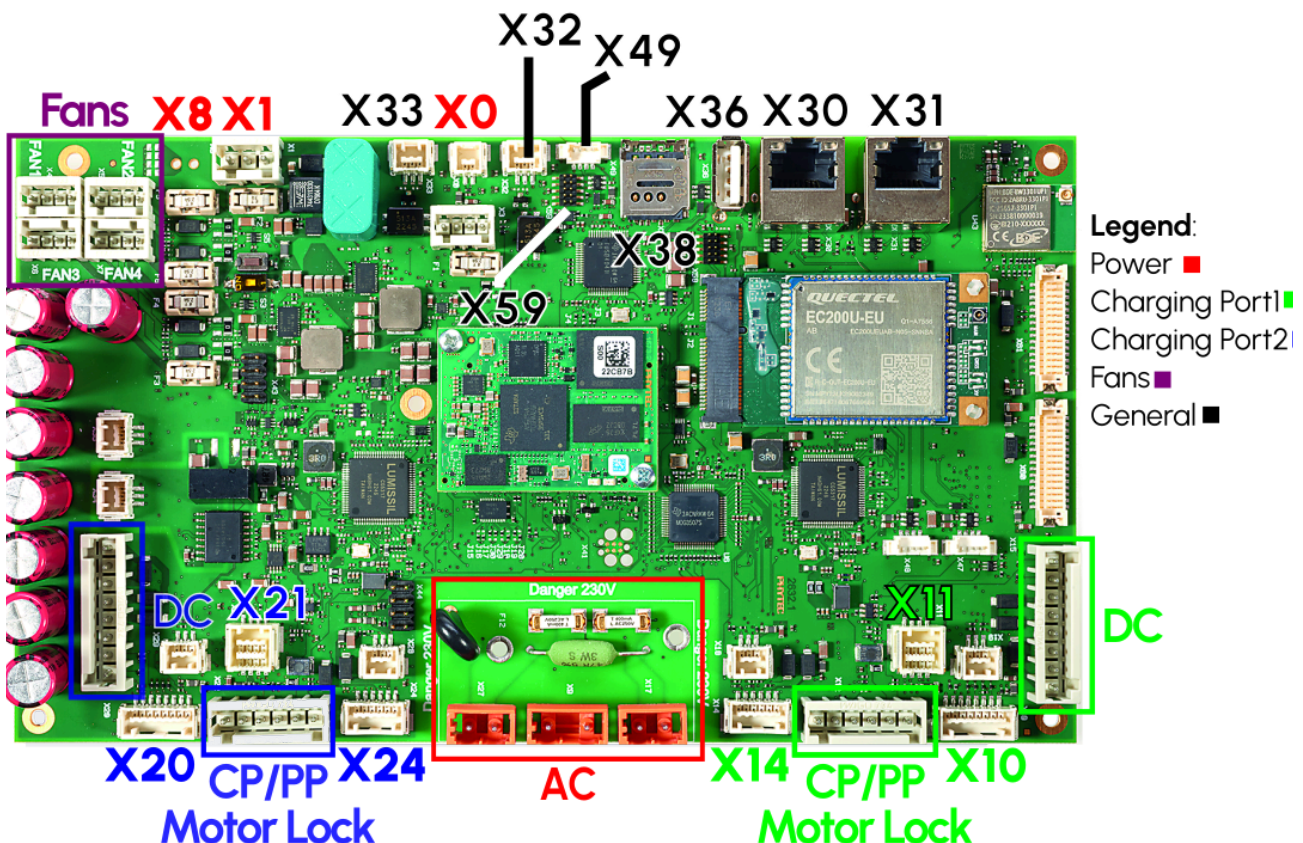
Electrical installations may be executed by qualified personnel only. High voltages / current conduction can be life endangering.

The complete hardware setup in the cabinet has to be evaluated for electrical and operational safety separately after installation.

Pionix can give no warranty on material or person damages.

Electrical installation

Connector location overview:



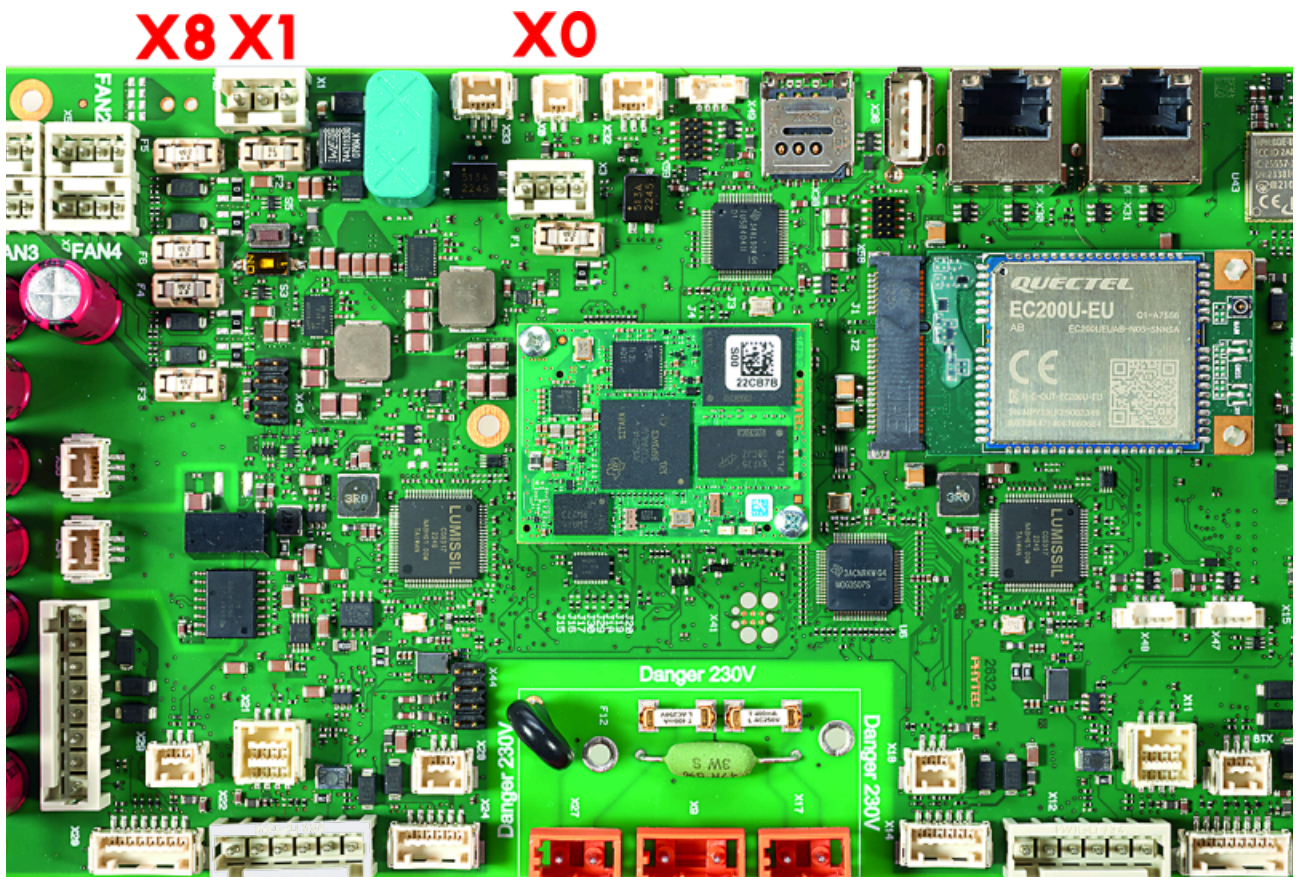
Preliminary requirements:

- Make sure that there is no voltage on the AC input cables during the complete installation procedure.

- Keep unqualified personnel at a safe distance during installation.
- Make sure that water cannot enter the cabinet.
- Protect the EVSE with safety devices and measures that the local rules specify.
- If it is necessary to remove safety devices, immediately install the safety devices after the work.
- Only use electrical wires of sufficient gauge and insulation to handle the rated current and voltage demand.
- Make sure that the load capacity of the grid is in accordance with the EVSE.
- Earth the EVSE correctly. Make sure that the EVSE is connected to a grounded, metal, permanent wiring system, or an equipment-grounding conductor must be run with the circuit conductors and connected to the equipment grounding terminal or lead on the product.
- Make sure that the connections to the EVSE comply with all applicable local rules.
- Make sure that the wiring inside the EVSE is protected from damage and cannot get trapped when you open or close the cabinet.

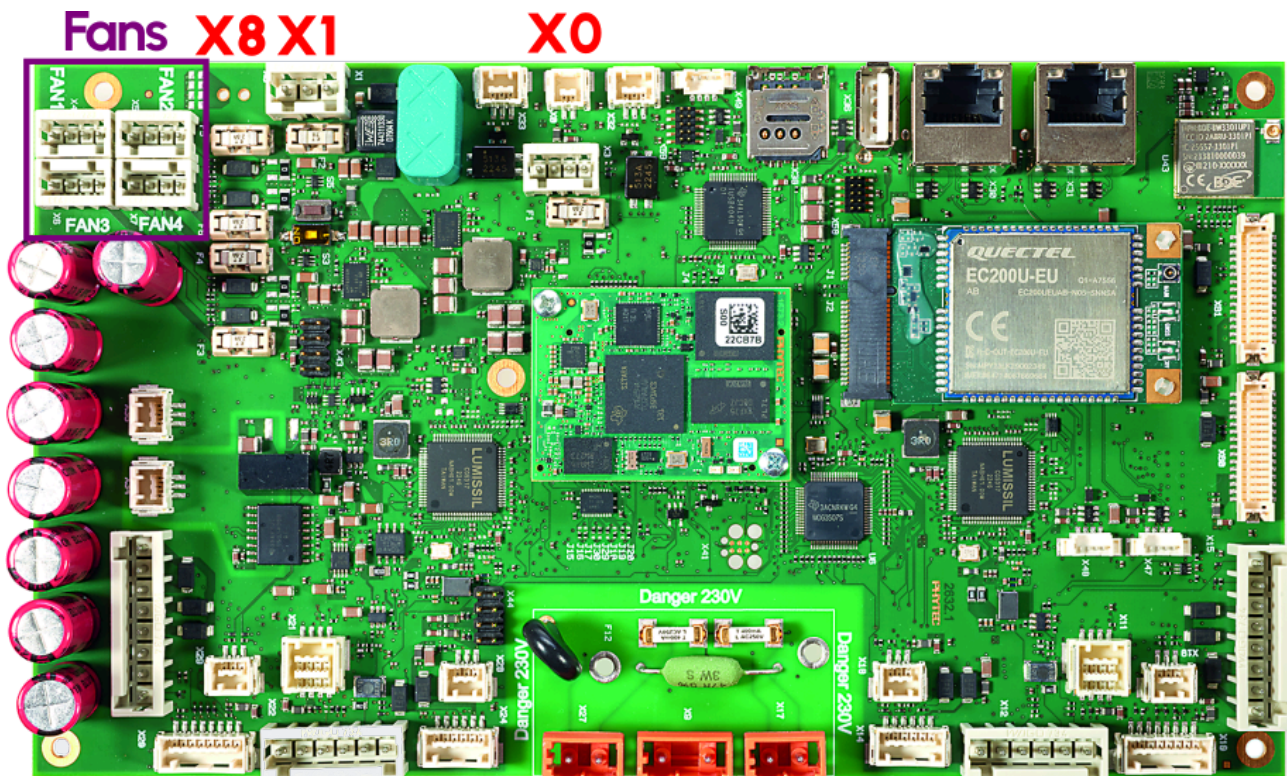
Connecting the phyVERSO-EVCS power supply:

The phyVERSO-EVCS is powered by a 12 V supply on the X1 connector with its pinout specified in the respective [Input supply](#) section. The location of this connector is highlighted in the picture below. **This connector is the central and only point where you should connect PE to the board.**



Setting up fans

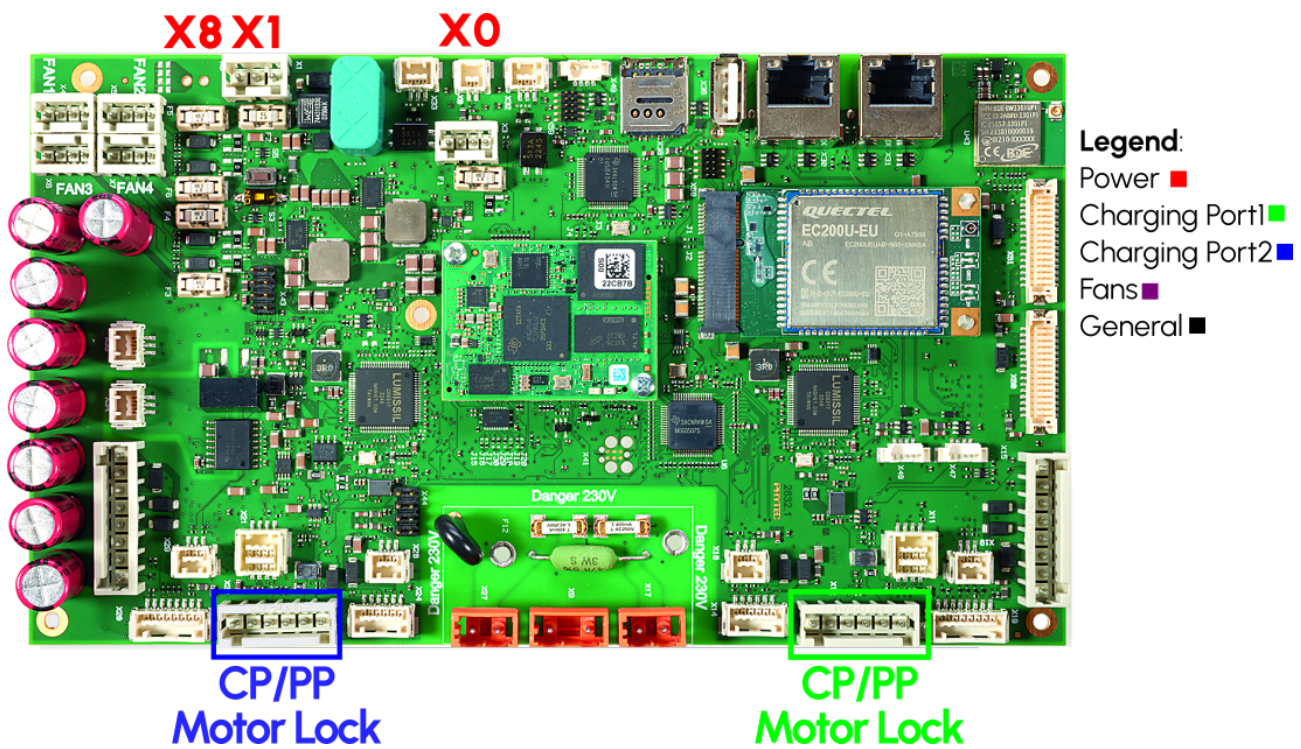
The controllable fans that are connected to the microcontroller and linux subsystems are available on X3-X7. For the pinout of the individual connectors, you can consult the [Fan connection](#) section further above.



If you want to attach an external fan supply, you will have to acquire a phyVERSO-EVCS hardware configuration that has the X8 connector populated. Get in touch with us or with PHYTEC for support on that matter.

Basic charging communication connection and motor locks

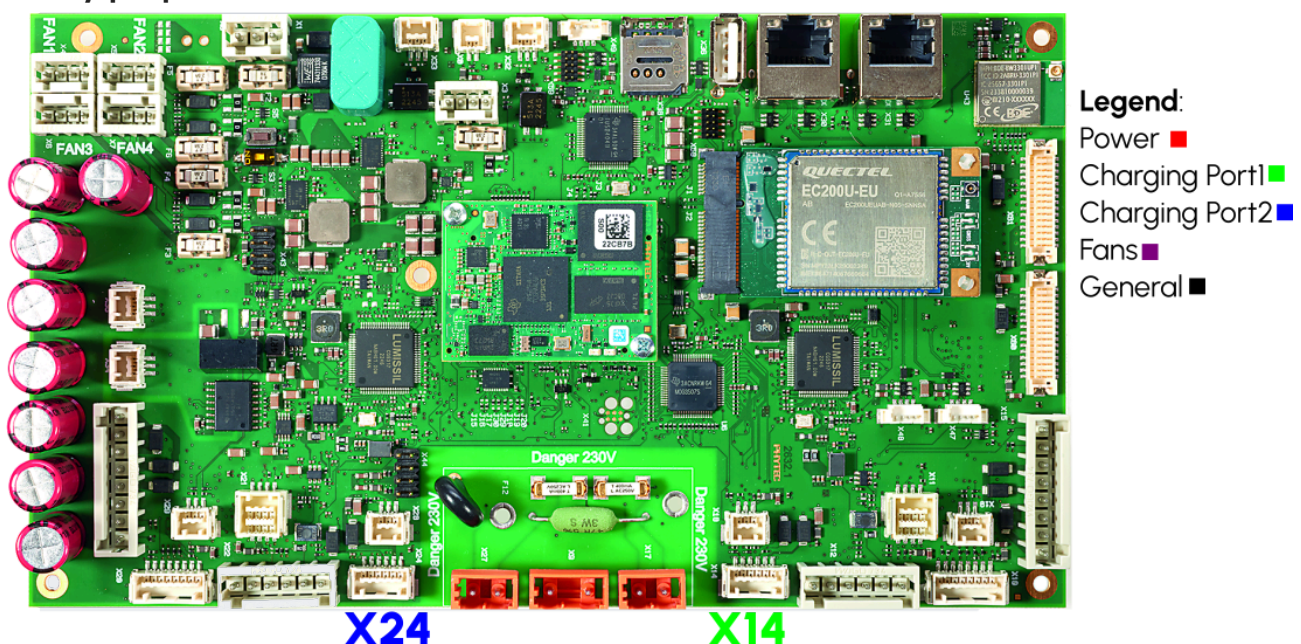
To set up the basic charging communication via CP/PP signaling you should connect your respective charging socket connections to the X12/X22 connectors for charging port1/port2. If your charging socket features a locking motor to ensure a locked charging connection, you can also connect the motor and feedback on these connectors. The pinouts are shown in the [Basic charging communication and motor lock connectors](#) section.



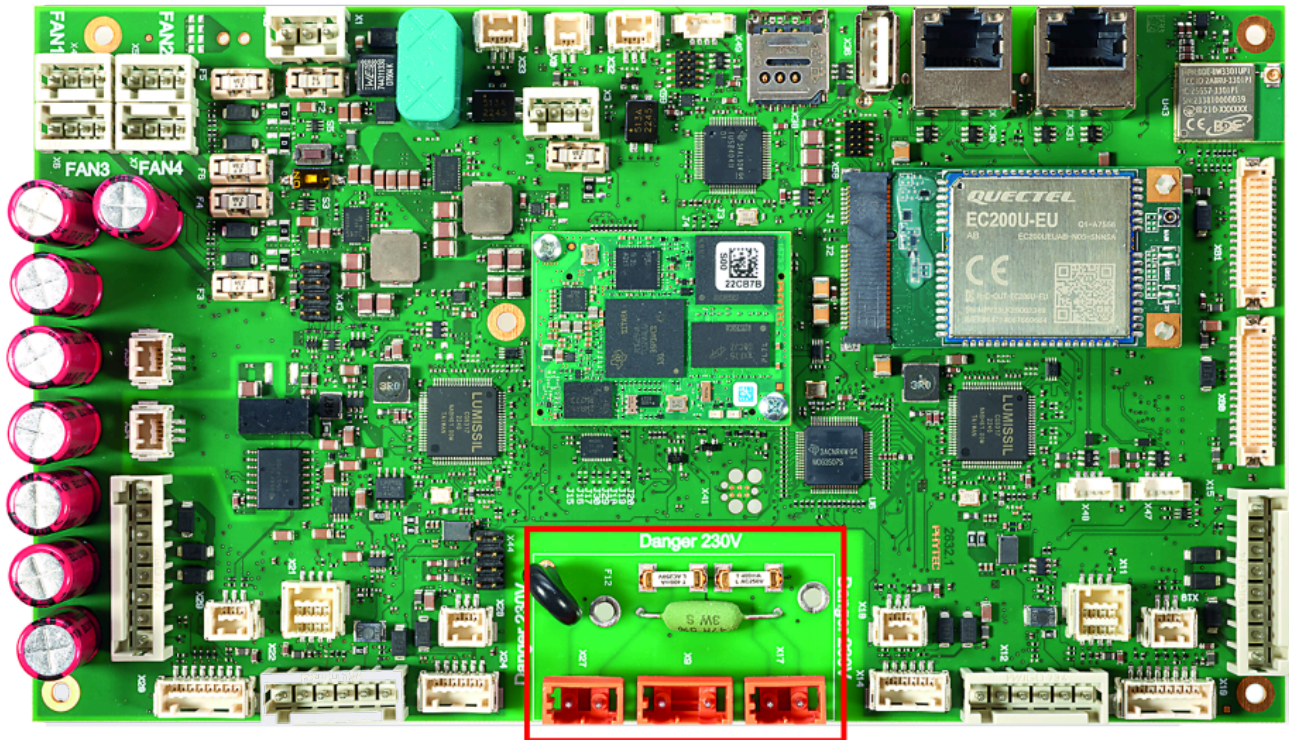
In most cases you should **NOT** attach PE at those connectors and keep the PE connection to board GND at X1 only

Setting up the RDC-MD connection

The separate RDC-MD for both charging ports are available on X14/X24. For pinouts and details on the additional NTC/PTC connections on those connectors consult the [Residual Direct Current Monitoring Device](#) section. **You will need to attach these for safety purposes!**



AC Charger Configuration

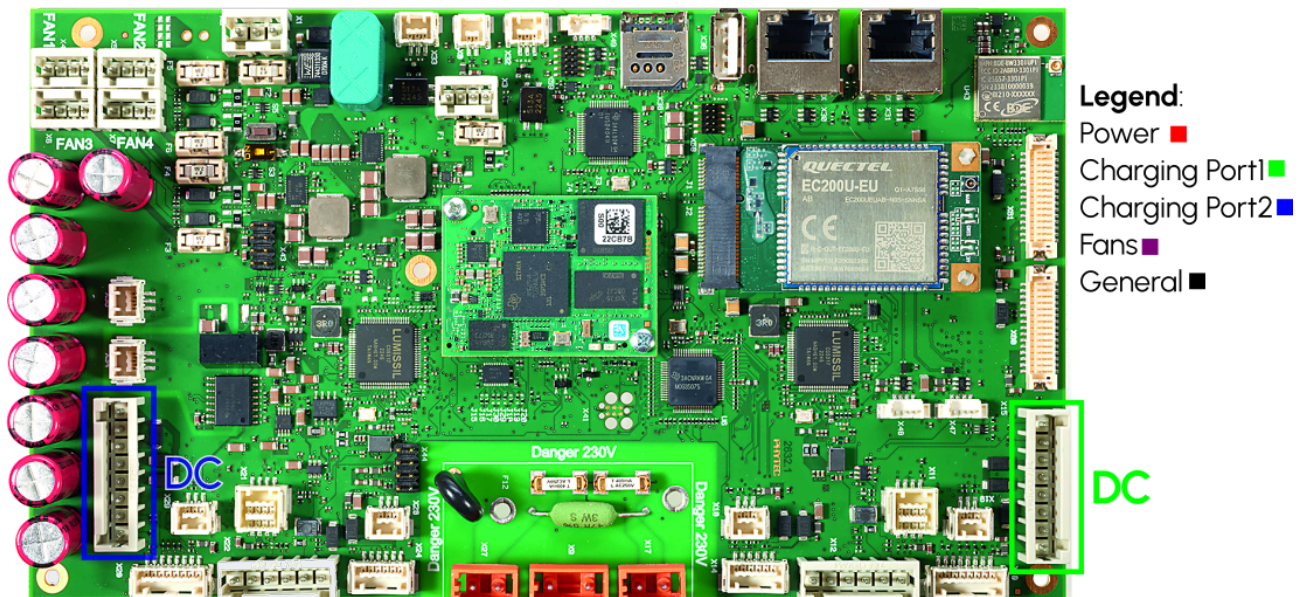


AC

To build an AC charger configuration on one or both of the charging ports you will first have to attach the AC supply at X9 to enable usage of PLC communication and AC contactor control/feedback. The pinout of this connector is shown in the [AC connection for PLC and AC contactor / feedback](#) section.

Afterwards, you will have to connect X17/X27 respectively to enable AC contactor control and feedback on the desired charging port. For pinout and details on said connectors please consult the [Control and mirror contacts for AC contactor](#) section.

DC Charger Configuration



To build a DC charger configuration on one or both of the charging ports you will have to connect your DC contactors and contact mirror feedback connections for your desired charging ports respectively (X15/X25). For pinout and details on said connectors, please consult the [Control and mirror contacts for DC contactor](#) section.



Caution

Please make sure to connect pin 1 to your DC coil 12 V / 24 V supply to use the integrated flyback diodes or provide external protection circuitry (e.g. flyback diode) against voltage spikes on pins 3/4/5 by inductive loads of the DC coils of your contactors.

In case of a DC configuration of a charging port, when not using an additional RDC-MD connected to this ports respective connector ([Residual Direct Current Monitoring Device](#)), please short the FAULT pin to GND. Otherwise this charging port will be in emergency stop mode.

Flashing/Updating the Linux image

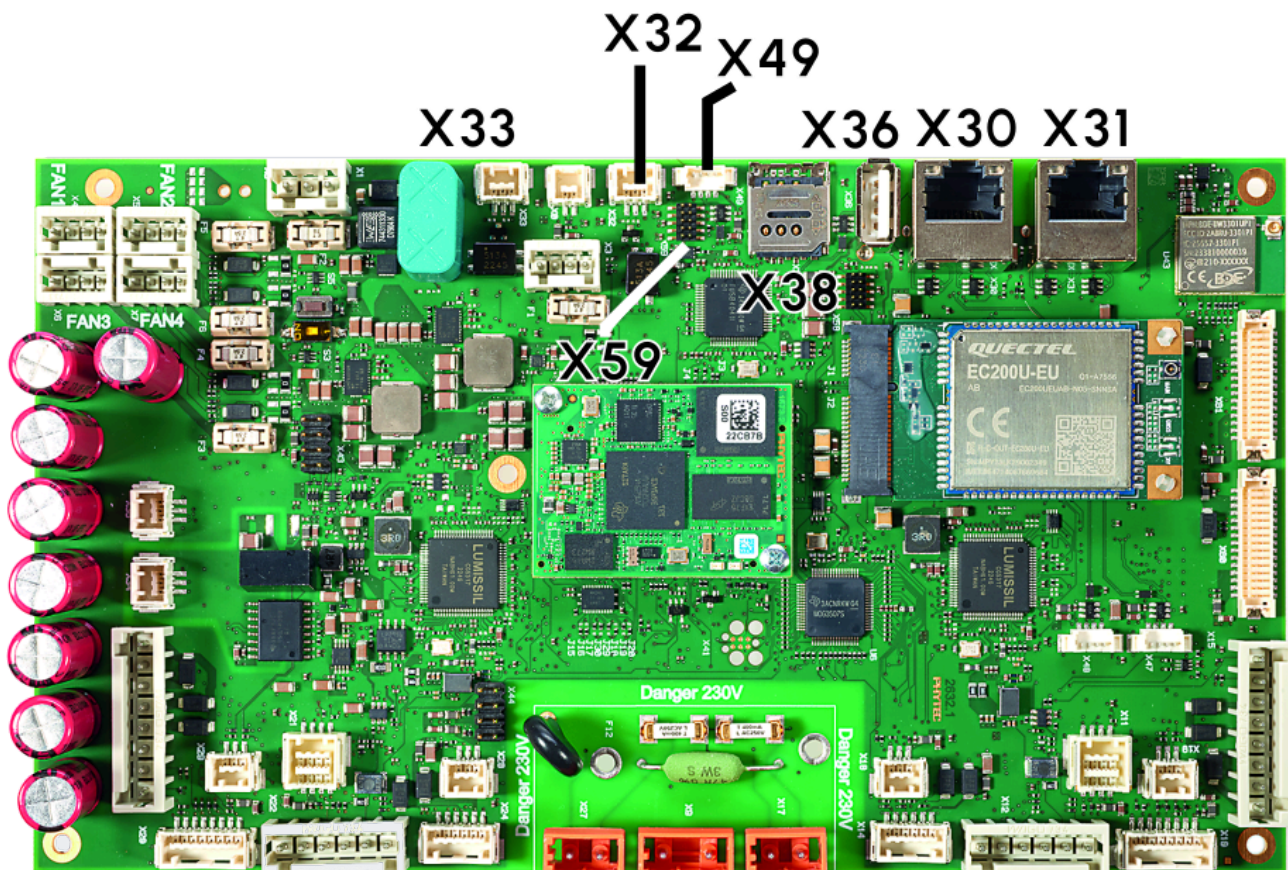
Flashing via RAUC

By default, the onboard SOM will be shipped to you containing a linux image that already supports RAUC updating. You will only have to access the phyVERSO-EVCS

via SSH ([Accessing the phyVERSO-EVCS via SSH](#)) and after login you can flash your desired image by executing the following command:

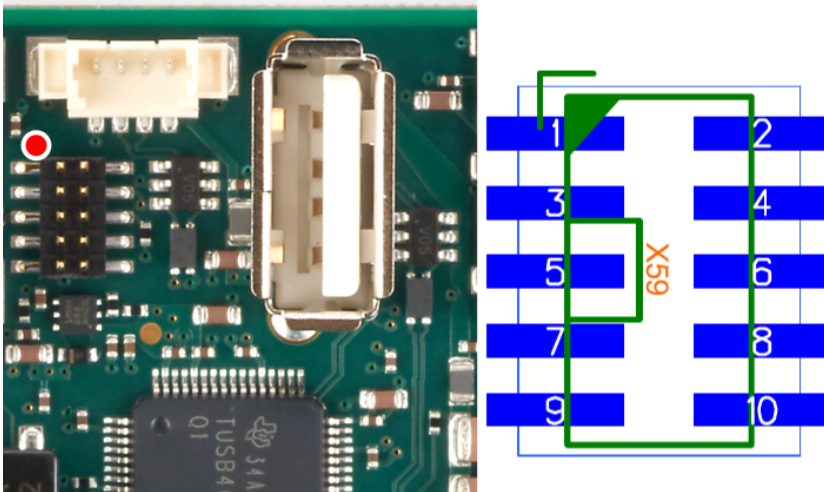
```
rauc install <IMAGE URL>
```

Flashing via u-boot



In case your boards SOM was not shipped with a RAUC supported image or something else goes wrong there is an emergency way to flash the SOM manually via USB over u-boot.

To access the u-boot shell, you will have to attach a 3.3 V serial connection to the X59 debug connector. More details can be found in the [Debugging](#) section.



Create only one partition <4 GB on the USB drive and format it with ext4. In the u-boot shell that is exposed on the debug UART on boot, the image file is copied in parts first from the USB drive into RAM and then from RAM to eMMC. Uncompress the image if necessary, e.g.:

```
unxz phyverso-everest-image-am62-ksp0728-1.wic.xz
```

Then copy the image file (.wic) onto the USB drive and insert it into the USB port X36 next to the debug serial connector. Connect to the serial console and power up the board. Press any key to stop auto boot. Then you should see the u-boot shell.

Now flash in the u-boot shell:

```
usb start  
run flash_emmc
```

If your file is not named "phyverso-basecamp-image-am62-ksp0728-1.wic", you can set a different file name before flashing:

```
setenv wic_image myfilename.wic
```

If you do not have the flash_emmc script, add it to the environment by copying this into u-boot shell:

None

```
setenv flash_emmc 'test -n ${BOOT_ORDER} || setenv
BOOT_ORDER system0 system1;test -n ${BOOT_system0_LEFT}
|| setenv BOOT_system0_LEFT 3;ext4size usb 0:1
${wic_image};setenv counter 0;setenv offset 0;setenv
block_number 0x2000;setexpr bytes_left ${filesize} -
${offset};echo start: ${bytes_left};while itest
${bytes_left} > 0;do echo rest: ${bytes_left};setexpr
copy_bytes ${block_number} * 0x200;ext4load usb 0:1
0xa0000000 ${wic_image} ${copy_bytes} ${offset};mmc write
0xa0000000 ${counter} ${block_number}; setexpr counter
${counter} + ${block_number};setexpr offset ${offset} +
${copy_bytes};echo ${counter};echo ${offset};if itest
${bytes_left} < ${copy_bytes}; then setenv bytes_left
0;else setexpr bytes_left ${bytes_left} -
${copy_bytes};fi;done;'
```

Once done, reset the board. It should now boot into the newly installed image.

If it does not find the root partition after boot, set up RAUC booting:

```
env default -a
setenv doraucboot 1
saveenv
```

The bootloader in the EMMC boot partition is not updated with the flash_emmc script. To write the latest boot loader into the emmc boot partition from Linux, use:

```
dd if=/boot/tiboot3.bin of=/dev/mmcblk0boot0 count=1024 conv=fsync
dd if=/boot/tispl.bin of=/dev/mmcblk0boot0 seek=1024 count=3072 conv=fsync
dd if=/boot/u-boot.img of=/dev/mmcblk0boot0 seek=5120 count=3072 conv=fsync
```

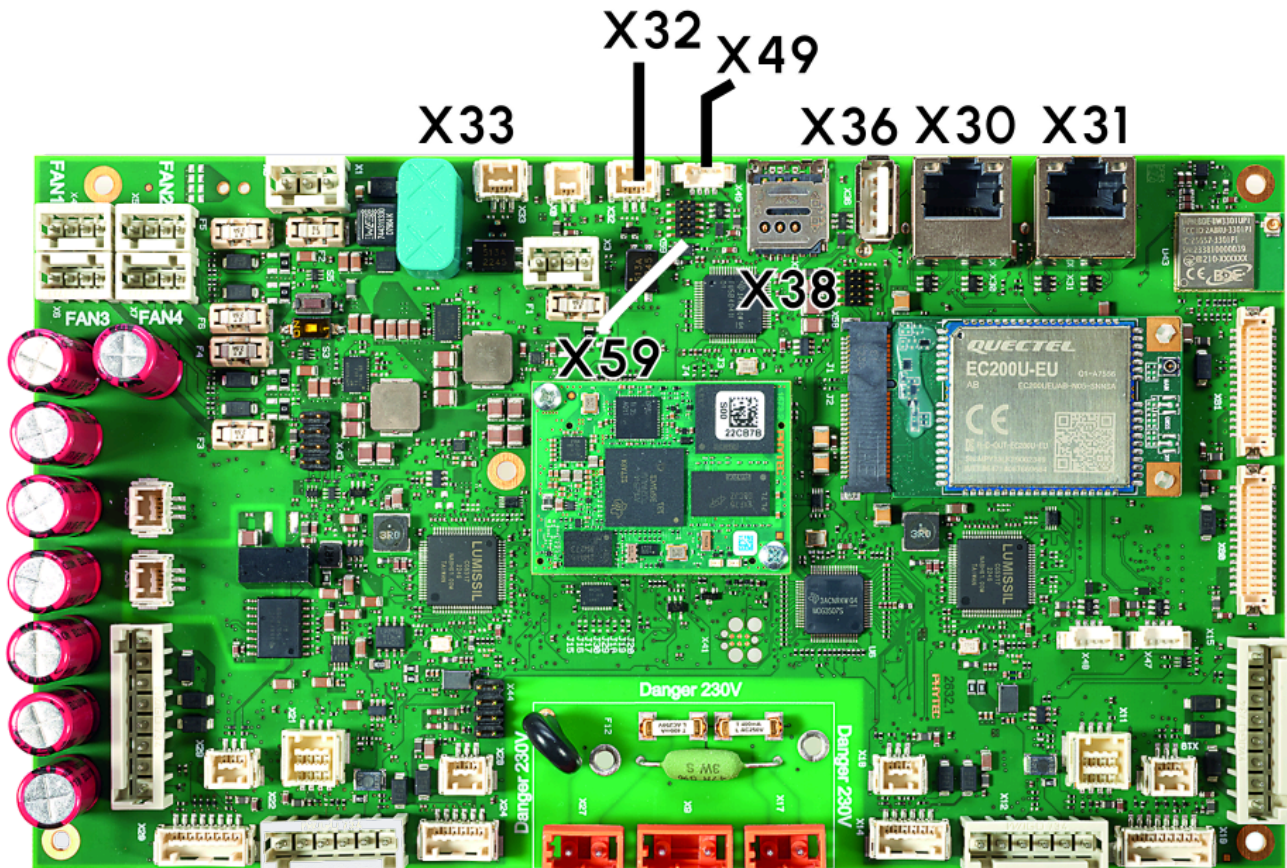
If you are using a blank new device, you will need to enable emmc boot partition boot first, this is only once per lifetime.

```
mmc bootpart enable 1 1 /dev/mmcblk0  
mmc bootbus set single_backward x1 x8 /dev/mmcblk0  
mmc hwreset enable /dev/mmcblk0
```

When flashing via USB, the system will fail to boot because the flashing process can not fully write the overlay partition on `/dev/mmcblk0p8`. On first boot into your new linux image you will be put into an emergency shell where you can format said partition and the system will boot normally afterwards (and after `exit`ing from the emergency shell). So execute the following from within the emergency shell:

```
mkfs.ext4 /dev/mmcblk0p8  
exit
```

Accessing the phyVERSO-EVCS via SSH



To access the phyVERSO-EVCS via SSH, you will have to attach a network cable to the RJ45 connector X30/X31. By default, the linux image exposes the network interface on X30 as "eth0" with the following static address configuration:

Parameter	Value
IP Address	192.168.3.11
Netmask	255.255.255.0

The network interface on X31 is exposed as "eth1" with the following static address configuration:

Parameter	Value
IP Address	192.168.4.11
Netmask	255.255.255.0

By default, the linux image allows SSH access on the root user with an empty password. So you can access the linux on X30 via:

```
ssh root@192.168.3.11
```

or on X31 via:

```
ssh root@192.168.4.11
```

Updating the microcontroller firmware

The most current version of the board contains an automatic update mechanism. The latest MCU firmware is part of the image and will automatically be flashed, if an older MCU FW version is detected during startup of the board.

Should you need to manually manage the firmware of the MCU, follow this process:



Caution: Backup firmware file before manual update

Please be aware that when you flash a firmware file manually it will get overwritten by the automatic MCU flashing process on the next boot. If you want to have the manually written firmware binary persistent across boots you will have to copy your firmware file to `/usr/lib/firmware/phyverso-firmware.bin`.

Also be aware that you are overwriting the firmware file that is provided by the image, so compatibility might be lost. Therefore, you should backup the firmware file before overwriting it or you will have to re-flash the whole linux image via RAUC to get the image-provided firmware file back in place.

The linux image of your phyVERSO-EVCS includes a tool to flash the microcontroller from within the linux system. Before trying to flash the microcontroller via UART from

within the linux system, make sure that no other process (e.g. everest or phyverso_cli) is currently using the serial `/dev/ttyS6`.

If provided with a firmware file by PIONIX or PHYTEC for update purposes, please copy the file to your phyVERSO-EVCS board via ssh/scp. On default, the linux image provides ssh access via root account and an empty password at 192.168.3.11 or 192.168.4.11 depending on which ethernet connector you are using to connect to the board (see [Accessing the phyVERSO-EVCS via SSH](#)). You will have to adjust user/pw and IP settings according to your setup if you have changed any default settings.

```
scp firmware.bin root@192.168.3.11:/root/
```

Assuming you transferred the binary to `/root/`, you can then flash the firmware using the `MSPM0_bsl_flasher` tool from within the linux system.

```
MSPM0_bsl_flasher flash -i firmware.bin -p /dev/ttyS6
```

Depending on the debug flags of the currently employed `MSPM0_bsl_flasher` version on target, you should see a similar output at the end of a successful flashing process:

```
Status:
  Programmed: 1
  Verified: 1
  Started: 1
```

The `MSPM0_bsl_flasher` tool automatically starts the application after successful flashing. You can also check if flashing was successful by running `phyverso_cli /dev/ttyS6` and check if you receive state updates.

In case of errors, try repeating the flashing process and make sure serial is not used by other processes. Also make sure that no other device is actively or passively in control of the nRST or BSL pins. It is a known problem that a XDS110 debugging

probe, that is attached to the uC target but not plugged into USB, is preventing a successful connection to the BSL device or any other startup.

BaseCamp Configuration

You can find more information about the configuration of PIONIX BaseCamp in the dedicated documentation. To access that documentation, use the credentials and the link which you received with your shipment.



Appendix

List of connectors

Reference	Function/Pinout	Connector
General		
X0	5 V Power output (5 V, GND)	Molex CLIK-Mate 502584-0260
X1	Input supply	WAGO MCS MINI 734-133

X3	SOM Fan	WAGO MCS MICRO 733-334
X4/X5/X6/X7	Fan connection	WAGO MCS MICRO 733-334
X8	Fan connection	WAGO MCS MINI 734-132
X9	AC connection for PLC and AC contactor / feedback	WAGO MCS MIDI Classic 231-332/001-000
Charging Port1/Port2		
X10/X20	Extension connectors	Molex CLIK-Mate 502584-0860
X11/X21	Extension connectors	Molex CLIK-Mate 503154-0870
X12/X22	Basic charging communication and motor lock connectors	Wago 734-136
X14/X24	Residual Direct Current Monitoring Device	Molex CLIK-Mate 502584-0660
X15/X25	Control and mirror contacts for DC contactor	Wago 734-138
X17/X27	Control and mirror contacts for AC contactor	WAGO MCS MIDI Classic 231-332/001-000
X18/X28	ADC inputs	Molex CLIK-Mate 502584-0360
X19/X29	ADC inputs	Molex CLIK-Mate 502584-0360
Interfaces		
X30/X31	Ethernet	RJ45
X32/X33	CAN	Molex CLIK-Mate 502584-0360
X34/X35	RS485	Molex CLIK-Mate 502584-0360

X36	USB	USB Type-A
X38	Mini-PCle/LTE	Micro-SIM Card Holder
X39	Mini-PCle/LTE	miniPCle-Connector
X41	Co-Processor Debug	TC2030-IDC
X43	Digital input and output	Header 2x4 2.54mm
X44	Digital input and output	Header 2x4 2.54mm
X47/X48/X49	USB	Molex 53398-0471
X58	M4 UART RX/TX (unused)	Header 2x5 1.27mm
X59	Debugging	Header 2x5 1.27mm
X60/X61	Display	DFI3EA-30DP-1.25V(52)

Documentation revision history

Date	Revision	Changes
2024-02-09	0	
2024-06-03	1	Adjusted documentation according to HW revision 1 changes
2024-10-08	2	Finalized version to be published.